Multicast Video Distribution: Protocol Design, Configuration, and Verification using PIM Sparse-Dense Mode

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Abstract

This article examines the design, configuration, and verification of IP multicast video distribution within an enterprise hotel network environment. Following an evaluation of multicast communication models, the Internet Group Management Protocol (IGMP) and Protocol Independent Multicast (PIM) variants are reviewed, including PIM Dense Mode, PIM Sparse Mode, PIM Sparse-Dense Mode, Source-Specific Multicast (SSM), and Bidirectional PIM (BIDIR-PIM). A prototype network is designed and configured using PIM Sparse-Dense Mode with multiple Rendezvous Points (RPs) employing Auto-RP for automatic RP discovery. Load balancing with automatic failover between primary and backup RPs is implemented and verified. Multicast traffic flow, routing table analysis, and failover activation are systematically verified using Cisco IOS diagnostic commands.

1. Introduction

In recent years, multimedia applications have become increasingly widespread across the internet, with a growing number of users engaging with this form of communication. IP multicasting has been successfully deployed in a variety of environments, including universities, schools, hospitals, and hotels. In hospital settings, live surgery can be observed by clinicians and students at multiple locations simultaneously. Hotels increasingly employ multicasting to deliver video streams and television channels to multiple rooms. Similarly, multicasting enables educational institutions to distribute software, lecture materials, video presentations, and live broadcasts to multiple users concurrently.

This article presents the design and implementation of a multicast video distribution network for GoodNight Plc, a hotel in the north of England, using PIM Sparse-Dense Mode with load-balancing failover via Auto-RP.

1.1 Multicast Video Communication

Multicasting is a communication model based on a one-to-many or many-to-many architecture, in which information or multicast packets are forwarded from one or multiple sources to several destinations. In a multicast network, a sender's data packets are transmitted over a single link that is distributed along the path to the intended set of destinations. The multicast video stream is replicated at points where distribution paths diverge, enabling delivery to targeted receivers on different network segments (Juyoung et al., 2001). Multicasting delivers an efficient multi-destination delivery system with optimised packet replication.

Without multicasting, unicast data packets must be explicitly sent to each interested receiver, resulting in unnecessary bandwidth consumption. A unicast data packet has both a single source and a single destination IP address, whereas a multicast packet has a single source address but carries a destination address shared by multiple receivers.

Advantages of Multicast Video Communication

Bandwidth Efficiency: Most network communication methods involving the transmission of data to multiple recipients benefit from the bandwidth efficiency provided by multicast networks. Applications include video conferencing, audio and video broadcasting, news feeds, video collaboration, and website caching.

Optimised Performance: Multicasting significantly optimises network performance by enabling a single multicast packet or stream to be sent once and replicated into multiple copies for interested receivers, thereby preserving network bandwidth and reducing traffic redundancy.

Scalability: Multicasting is suited to both small and large environments, particularly WAN deployments. It enables video, live presentations, and news feeds to be distributed to multiple locations without consuming excessive bandwidth or broadcasting streaming files to all network clients.

Support for Distributed Applications: Multicast networks support a wide range of multimedia applications, including video conferencing, e-learning, database replication, and video broadcasts.

Network Economy and Availability: Through the management of network resources such as multicast traffic, CPU load, and server utilisation, network costs are reduced and susceptibility to congestion is minimised.

Challenges of Multicast Video Communication

Best-Effort Delivery: Multicast networks are based on UDP (User Datagram Protocol), in which packets are forwarded on a best-effort basis. This method does not guarantee reliable delivery to all destinations, and ongoing research continues to address this limitation.

Out-of-Sequence Packets: Since multicast transmission is inherently unreliable, packets may arrive out of sequence as a result of changes in network topology. Unlike TCP, which supports sequencing and acknowledgement, UDP multicast provides no such guarantees.

Congestion: In the absence of TCP windowing, flow control, or congestion avoidance mechanisms, real-time applications such as video or VoIP require Quality of Service (QoS) provisioning.

Duplicates: Certain routing protocol mechanisms, including Shortest Path Trees (SPTs) and register messages, may occasionally replicate multicast packets. Multicast applications should be designed to tolerate and handle such duplication.

2. Internet Group Management Protocol (IGMP)

The Internet Group Management Protocol (IGMP) is a protocol that manages the joining and leaving of multicast groups within a network. It can also be defined as a protocol that automatically manages the flow of multicast data traffic through a network using report messages and queries. IGMP is an essential component of the Internet Protocol suite and currently has three versions. It facilitates the registration of multicast group membership for hosts directly connected to a router.

IGMP operation is as follows: when a host joins a multicast group, it transmits a membership report message, which is received by the router. The router periodically sends queries to verify that hosts remain members of a multicast group. Hosts that remain members respond to the query with a membership report; in cases where multiple members exist, only one host responds. If the router issues a query and receives no response for a multicast group, it assumes that no hosts remain in the group and ceases sending packets to that network. IGMPv2 is the most commonly deployed version.

IGMP enables a host to inform the router of the group membership details of the hosts within the multicast group (Doyle and Carroll, 2001). By default, an IGMP query is forwarded by a router to all hosts in a network every 125 seconds, with the router destination address set to 224.0.0.1 for all host groups. The hosts in the group then respond with an IGMP membership report.

3. Protocol Independent Multicast (PIM)

Protocol Independent Multicast (PIM) was created by the Inter-Domain Multicasting Routing (IDMR) working group of the Internet Engineering Task Force (IETF). It is a collection of multicast routing protocols responsible for transmitting multicast data to destination addresses or multicast groups and for constructing multicast trees, either shared or source-based, and pruning those trees to prevent unnecessary packet forwarding. Although PIM is a multicast routing protocol, it uses the unicast routing table to execute Reverse Path Forwarding (RPF) rather than maintaining an entirely independent multicast routing table. All PIM protocols share a common control message format and depend on unicast routing information.

The three main PIM protocol variants are PIM Dense Mode (PIM-DM), PIM Sparse Mode (PIM-SM), and PIM Sparse-Dense Mode (PIM-SM-DM). Bidirectional PIM (BIDIR-PIM) and PIM Source-Specific Multicast (PIM-SSM) are also available but are less commonly deployed than PIM-SM, PIM-DM, or PIM-SM-DM.

3.1 PIM Dense Mode

Protocol Independent Multicast Dense Mode (PIM-DM) is a multicast routing protocol used to flood multicast traffic across all segments of a network. It uses Reverse Path Forwarding (RPF) to forward traffic away from sources. Each multicast delivery tree originates at the source, ensuring that traffic takes the shortest path from source to destination. Multicast routers with no downstream receivers transmit a prune message to halt the multicast traffic flow (Jelger and Noel, 2008). This flooding mechanism is re-activated every three minutes in case multicast branches subsequently require the data.

PIM Dense Mode is primarily suitable for densely distributed networks or distributions in which active receivers exist on every subnet. This method distributes traffic to all points in a network, potentially resulting in bandwidth wastage and increased CPU processing load. Routers gather information through the prune-and-

flood mechanism, and multicast traffic entries are identified using (S, G) notation, where S is the source address and G is the group address.

3.2 PIM Sparse Mode

Protocol Independent Multicast Sparse Mode (PIM-SM) is a multicast routing protocol designed for multicast traffic destined for receivers distributed sparsely across a wide geographical area. This model uses a shared tree delivery approach for each multicast group rather than the source tree used by PIM Dense Mode. It is assumed that receivers must explicitly join a multicast group to receive multicast data. This means that multicast packets are transmitted only when explicitly requested (Handley et al., 2001).

The Rendezvous Point (RP) functions as the meeting point for both multicast sources and receivers in PIM Sparse-Dense Mode. Every multicast group is assigned an RP, which is used to forward all multicast traffic destined for that group. RPs enable multiple multicast sources to share the same multicast delivery tree. Although PIM Sparse Mode is based on a shared delivery tree, it can also implement a Shortest Path Tree for a multicast receiver. To forward data to a specified multicast address, the source station must register with the RP, after which the packet is replicated and forwarded through the shared tree route to the multicast receiver (Williamson, 1999).

3.3 PIM Sparse-Dense Mode

PIM Sparse-Dense Mode combines the characteristics of both Sparse Mode and Dense Mode. It permits an interface to operate in either Sparse Mode or Dense Mode depending on the application or multicast group (Doyle and Carroll, 2001). If a multicast group is configured with an RP or the interface receives (*, G) notation, it operates in Sparse Mode. If no RP is configured or the interface receives (S, G) notation, it operates in Dense Mode.

A principal advantage of PIM Sparse-Dense Mode is the elimination of manual RP configuration on every router in the network. The Auto-RP model, which is managed by Dense Mode and SPTs, automatically selects RPs, significantly reducing administrative overhead.

3.4 Source-Specific Multicast (SSM)

Source-Specific Multicast (SSM), defined in RFC 4607, is a multicast routing protocol in which multicast forwarding is determined not only by the multicast group address but also by the source address. An SSM group address is denoted by (S, G), where S represents the source address and G represents the group address, compared to Sparse Mode which uses (*, G) notation. The multicast address range 232.0.0.0/8 has been reserved by the IANA for SSM in IPv4, and FF3X::/32 in IPv6.

Unlike PIM Sparse Mode, in which the RP establishes a shared multicast delivery tree (Cisco, 2008), SSM identifies the source of a multicast channel in advance. The multicast delivery tree is therefore established directly from the channel host to the source, eliminating the need for an RP to link the source to the shared tree.

3.5 Bidirectional PIM (BIDIR-PIM)

Bidirectional PIM is a multicast routing protocol designed to support financial and communication applications that require many-to-many communication. It supports applications with multiple sources and receivers. In PIM Sparse Mode, source trees connect active sources to the RP through a registration process. In BIDIR-PIM, source trees are replaced by a mechanism known as the Designated Forwarder (DF), which determines which multicast packets should be forwarded to the RP (Cisco, 2008). In environments where sources are also

receivers, multicast packets are forwarded both from the RP to the receivers and from the sources to the RP, creating a bidirectional shared tree channel.

3.6 Role of Unicasting within a Multicast Network

In multicast networks, routing is based on the transmission of multicast packets from the source towards the distribution tree and onwards to interested receivers (Parkhurst, 1999). To prevent multicast packets from being forwarded to incorrect destinations or creating routing loops, a mechanism known as Reverse Path Forwarding (RPF) is employed. RPF is a logical mechanism that roots multicast forwarding from the receiver back to the shared or source tree. A multicast routing protocol uses information from the unicast routing table to execute RPF, determining the closest interface to the multicast source rather than constructing an entirely independent multicast routing table.

3.7 Session Announcement Protocol (SAP)

Session Announcement Protocol (SAP) is an announcement protocol used to broadcast multicast conference sessions and to disseminate session setup information to potential multicast participants via their multicast address and port. Session announcement packets are multicast to potential participants and contain the session description and an authentication header carried using UDP.

A SAP listener learns all SAP addresses and ports within its scope and, over time, learns all multicast sessions being announced, allowing these sessions to be aggregated. SAP is used to announce video streams on a multicast video network as predefined by Time to Live (TTL) or multicast group address range (Handley et al., 2001).

4. Configuring PIM Sparse-Dense Mode

4.1 Enabling PIM Sparse-Dense Mode on a Router

To enable PIM Sparse-Dense Mode on a router, all interfaces must be configured with the ip pim sparse-dense-mode command in the Cisco IOS interface configuration mode. The following example uses Router 3 (R3) from the prototype network topology:

```
R3# configure terminal
Enter configuration commands, one per line. End with CONTROL/Z
R3(config)# ip multicast-routing
R3(config)# interface GigabitEthernet0/0
R3(config-if)# ip pim sparse-dense-mode
R3(config)# interface GigabitEthernet0/1
R3(config-if)# ip pim sparse-dense-mode
R3(config)# interface GigabitEthernet0/2
R3(config-if)# ip pim sparse-dense-mode
```

4.2 Load Balancing with Failover Between Multiple RPs Using Auto-RP

Once PIM Sparse-Dense Mode has been configured on the multicast network, the Rendezvous Point (RP) must be configured. RPs can be configured either manually or through RP discovery protocols. Protocol Independent Multicast networks support two principal RP discovery protocols: Auto-RP and Bootstrap Router

(BSR). For this network topology, multiple RPs are configured using Auto-RP with load balancing and failover enabled.

To enable Auto-RP, Candidate Rendezvous Points (C-RPs) and an RP Mapping Agent must be configured. In this topology, Router 1 (R1) acts as the Mapping Agent while Routers 3 (R3) and 4 (R4) act as Candidate RPs configured on loopback interfaces. Two loopback interfaces per router are required to support load balancing with failover.

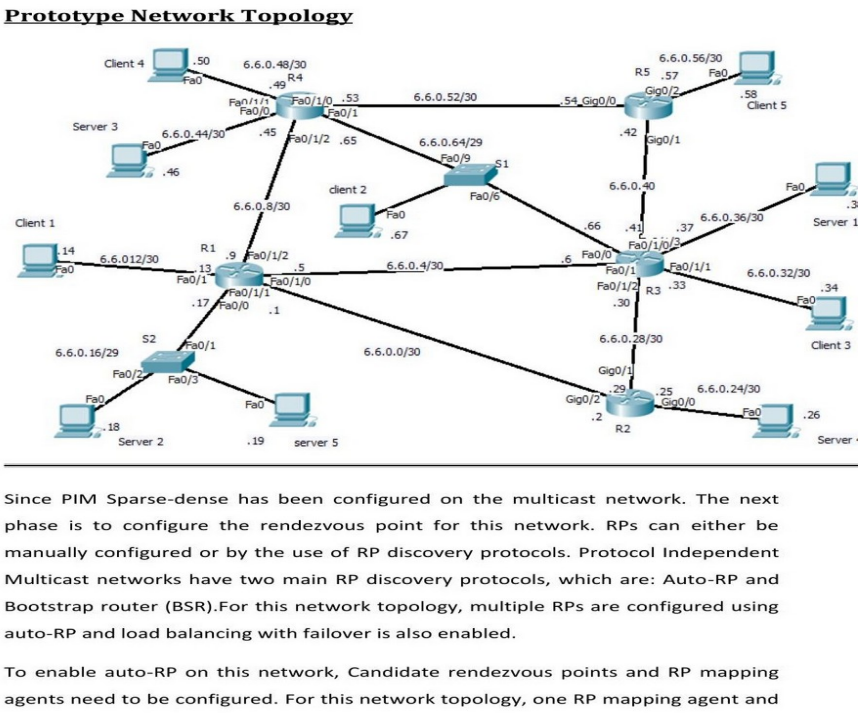


Figure 1: Prototype Network Topology

Candidate RP Configuration on R3

```
R3(config)# interface loopback 0
R3(config-if)# ip address 6.6.1.5 255.255.255.255
R3(config-if)# ip pim sparse-dense-mode
R3(config-if)# exit
R3(config)# interface loopback 1
R3(config-if)# ip address 6.6.1.2 255.255.255.255
R3(config-if)# ip pim sparse-dense-mode
R3(config-if)# exit
R3(config)# ip pim send-rp-announce lo0 scope 6 group-list 2 interval 10
R3(config)# ip pim send-rp-announce lo1 scope 6 group-list 1 interval 10
R3(config)# access-list 1 permit 227.1.6.5
R3(config)# access-list 1 permit 227.1.6.1
R3(config)# access-list 1 permit 227.1.6.2
R3(config)# access-list 2 permit 227.1.6.3
R3(config)# access-list 2 permit 227.1.6.4
R3(config)# access-list 2 permit 224.2.127.254
```

Candidate RP Configuration on R4

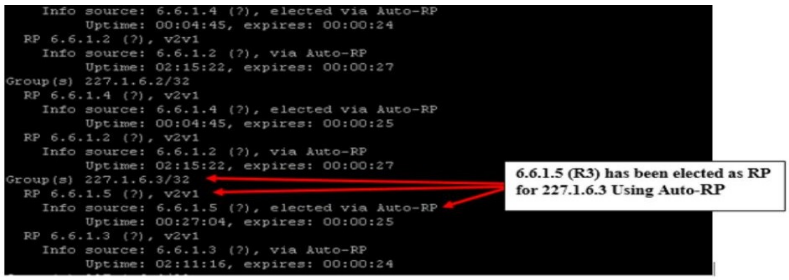
```
R4(config)# interface loopback 0
R4(config-if)# ip address 6.6.1.4 255.255.255.255
R4(config-if)# ip pim sparse-dense-mode
R4(config-if)# exit
R4(config)# interface loopback 1
R4(config-if)# ip address 6.6.1.3 255.255.255.255
R4(config-if)# ip pim sparse-dense-mode
R4(config-if)# exit
R4(config)# ip pim send-rp-announce lo0 scope 6 group-list 1 interval 10
R4(config)# ip pim send-rp-announce lo1 scope 6 group-list 2 interval 10
R4(config)# access-list 1 permit 227.1.6.5
R4(config)# access-list 1 permit 227.1.6.1
R4(config)# access-list 1 permit 227.1.6.2
R4(config)# access-list 2 permit 227.1.6.3
R4(config)# access-list 2 permit 227.1.6.4
R4(config)# access-list 2 permit 224.2.127.254
```

RP Mapping Agent Configuration on R1

```
R1(config)# interface loopback0
R1(config-if)# ip address 6.6.1.7 255.255.255.255
R1(config-if)# ip pim sparse-dense-mode
R1(config)# ip pim send-rp-discovery loopback0 scope 6 interval 10
R1(config)# exit
```

At this stage, all routers in the topology learn RP details from the Mapping Agent (R1), as all Cisco routers share the same RP-Discovery group address 224.0.1.40. Entering the Show ip pim rp mapping command on R1 verifies that Auto-RP is operational.

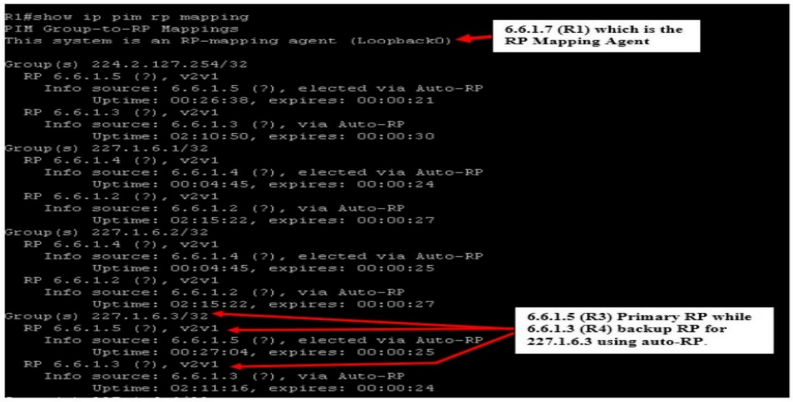
Multicast streams 227.1.6.5, 227.1.6.1, and 227.1.6.2 have their primary RP on Loopback 0, Group-List 1 on R4, with backup on Loopback 1 on R3. Multicast streams 227.1.6.3, 227.1.6.4, and 224.2.127.254 (SAP Stream) have their primary RP on Loopback 0, Group-List 2 on R3, with backup on Loopback 1 on R4.



Multicast streams 227.1.6.5, 227.1.6.1, 227.1.6.2 has their primary RP on loopback 0, group-list 1 on R4 and its backup on loopback 1 on R3 while multicast Stream **227.1.6.3**, 227.1.6.4 and 224.2.127 (SAP Stream) has its primary RP on loopback 0, group-list 2 on R3 and its backup on loopback 1 on R4.

Figure 2: Show ip pim rp mapping - First RP Mapping Verification (R1)

Show ip pim rp mapping



4.3 Comparison between the Operation of Auto-RP and BSR

Auto-RP and Bootstrap routers (BSR) are both RP discovery protocols which are mainly used Protocol Independent Multicast (PIM). With the use of both Auto-rp and Bootstrap routers (BSR) the need to manually configured rendezvous points (RPs) on every router is eliminated. Also in both auto-RP and bootstrap routers (BSR) multiple routers may be configured to serve as a failover or redundancy (Doyle and Carroll, 2001).

But unlike Auto-RP that uses multicast to send Rendezvous Point to multicast group

Figure 3: Show ip pim rp mapping - Second Verification Showing Elected RP

4.3 Comparison Between Auto-RP and Bootstrap Router (BSR)

Auto-RP and Bootstrap Router (BSR) are both RP discovery protocols used within Protocol Independent Multicast networks. Both protocols eliminate the need to manually configure RPs on every router, and both support multiple RPs configured as failover or redundancy nodes (Doyle and Carroll, 2001).

Unlike Auto-RP, which uses multicast to distribute RP-to-group mapping information, BSR floods its BSR messages (sent by the current Candidate-BSR) to all groups throughout the network. As a result of how RP information is distributed in BSR, TTL scoping cannot be utilised, and during PIM Sparse-Dense operation, BSR does not revert to PIM Dense Mode. Additionally, unlike Auto-RP, announcements for Candidate RPs in BSR are transmitted to elected routers via unicast rather than multicast.

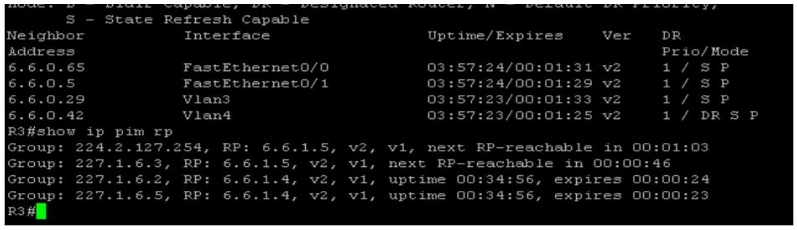
5. Verification of Multicast Communication

5.1 Verification of Multicast Traffic Flow

To verify that multicast traffic is flowing correctly from source to destination, two Cisco IOS show commands are used: Show ip pim neighbor and Show ip pim interface.

Show ip pim neighbor

This command displays the PIM Neighbour Table, listing all neighbour addresses participating in PIM, the interfaces through which they are connected to R3, and the uptime and timeout sessions for each interface. This command is used to verify that all PIM neighbours are operational, that interface connections are established, and that all interfaces are operating in the correct PIM mode.



```
      S - State Refresh Capable
Neighbor          Interface                Uptime/Expires    Ver   DR
Address                                                            Prio/Mode
6.6.0.65          FastEthernet0/0          03:57:24/00:01:31 v2    1 / S P
6.6.0.5           FastEthernet0/1          03:57:24/00:01:29 v2    1 / S P
6.6.0.29          Vlan3                    03:57:23/00:01:33 v2    1 / S P
6.6.0.42          Vlan4                    03:57:23/00:01:25 v2    1 / DR S P
R3#show ip pim rp
Group: 224.2.127.254, RP: 6.6.1.5, v2, v1, next RP-reachable in 00:01:03
Group: 227.1.6.3, RP: 6.6.1.5, v2, v1, next RP-reachable in 00:00:46
Group: 227.1.6.2, RP: 6.6.1.4, v2, v1, uptime 00:34:56, expires 00:00:24
Group: 227.1.6.5, RP: 6.6.1.4, v2, v1, uptime 00:34:56, expires 00:00:23
R3#
```

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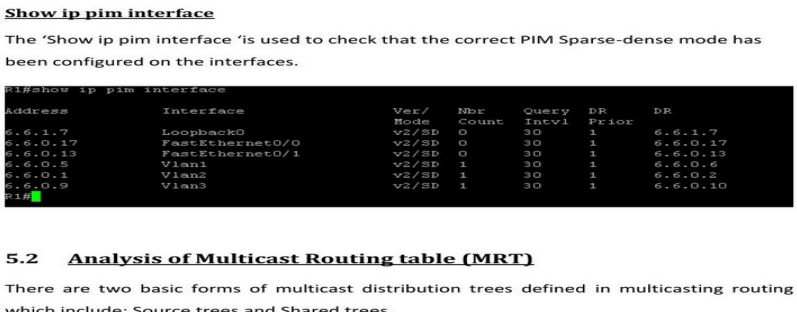
Figure 4: Show ip pim neighbor - PIM Neighbour Table

Show ip pim interface

The Show ip pim interface command is used to verify that the correct PIM Sparse-Dense Mode has been configured on all interfaces.



Show ip pim interface

The 'Show ip pim interface 'is used to check that the correct PIM Sparse-dense mode has been configured on the interfaces.

```
R1#show ip pim interface

Address          Interface                Ver/   Nbr    Query  DR     DR
                                          Mode   Count  Intvl  Prior
6.6.1.7          Loopback0                v2/SD  0      30     1      6.6.1.7
6.6.0.17         FastEthernet0/0          v2/SD  0      30     1      6.6.0.17
6.6.0.13         FastEthernet0/1          v2/SD  0      30     1      6.6.0.13
6.6.0.5          Vlan1                    v2/SD  1      30     1      6.6.0.6
6.6.0.1          Vlan2                    v2/SD  1      30     1      6.6.0.2
6.6.0.9          Vlan3                    v2/SD  1      30     1      6.6.0.10
R1#
```

5.2 Analysis of Multicast Routing table (MRT)

There are two basic forms of multicast distribution trees defined in multicasting routing which include: Source trees and Shared trees

Figure 5: Show ip pim interface - PIM Interface Verification

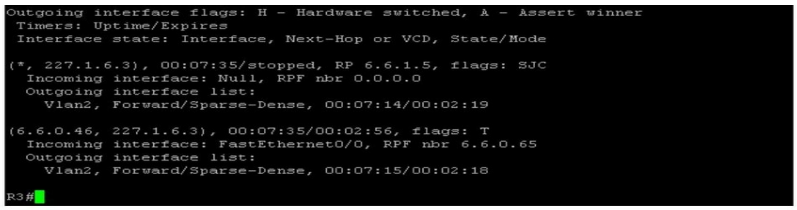
5.2 Analysis of the Multicast Routing Table

Two basic forms of multicast distribution trees exist within multicast routing: Source Trees and Shared Trees.

Source Trees: This distribution model roots the multicast tree at the source, linking it to all interested receivers in the multicast group. The path used to deliver multicast packets from the source to the targeted receivers is called the Shortest Path Tree (SPT), denoted by (S, G), where S is the source address and G is the group address.

Shared Trees: A shared tree is shared among multiple sources for a particular multicast group address. Unlike source trees, shared trees are rooted at a central point in the network called the Shared Root, denoted by (*, G), where the asterisk represents any source. Entering the command Show ip mroute 227.1.6.3 displays the information for multicast stream 227.1.6.3 in the multicast routing table.

From the output shown in Figure 6, the entry (*, 227.1.6.3) uses (*, G) notation where the asterisk represents any source and G is multicast group 227.1.6.3. The RP is identified as 6.6.1.5, the route was established 7 minutes 35 seconds ago, and the entry is stopped. The 'Null' incoming interface indicates that R3, which is also the RP, represents the closest interface.



```
Outgoing interface flags: H - Hardware switched, A - Assert winner
 Timers: Uptime/Expires
 Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 227.1.6.3), 00:07:35/stopped, RP 6.6.1.5, flags: SJC
  Incoming interface: Null, RPF nbr 0.0.0.0
  Outgoing interface list:
    Vlan2, Forward/Sparse-Dense, 00:07:14/00:02:19

(6.6.0.46, 227.1.6.3), 00:07:35/00:02:56, flags: T
  Incoming interface: FastEthernet0/0, RPF nbr 6.6.0.65
  Outgoing interface list:
    Vlan2, Forward/Sparse-Dense, 00:07:15/00:02:18

R3#
```

Entering the 'Show ip mroute 'command displays the route for all the grouped multicast since the specific multicast stream address wasn't specified.

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Figure 6: Show ip mroute 227.1.6.3 - Specific Multicast Route Entry

The entry (6.6.0.46, 227.1.6.3) uses (S, G) notation with source address 6.6.0.46 and multicast group address 227.1.6.3. The uptime is 7 minutes 35 seconds, with an expiry time of 2 minutes 56 seconds. The incoming

interface is FastEthernet0/0 (the closest interface to the source), identified as RPF 6.6.0.65. The outgoing interface, Vlan 2, defines how multicast traffic for this group is forwarded.

Entering Show ip mroute without a specific address displays routing entries for all multicast groups.

Show ip mroute

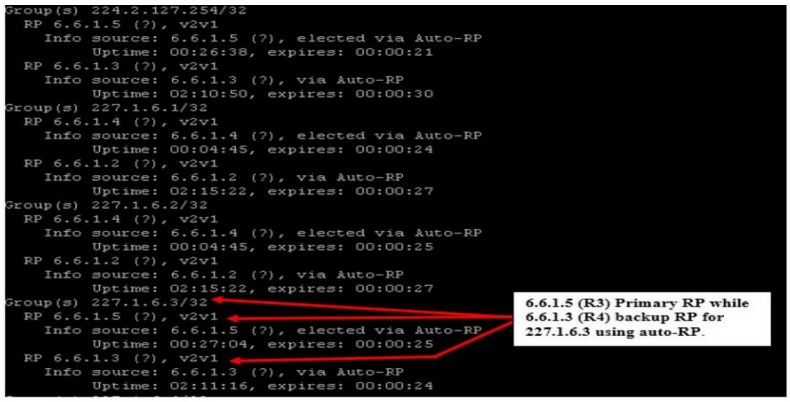
```
(6.6.0.38, 227.1.6.1), 00:43:37/00:03:23, flags: FT
  Incoming interface: Vlan1, RPF nbr 0.0.0.0
  Outgoing interface list:
    FastEthernet0/1, Forward/Sparse-Dense, 00:42:03/00:02:45

(*, 227.1.6.3), 00:58:29/stopped, RP 6.6.1.3, flags: SJC
  Incoming interface: FastEthernet0/0, RPF nbr 6.6.0.65
  Outgoing interface list:
    Vlan2, Forward/Sparse-Dense, 00:58:07/00:02:22

(6.6.0.46, 227.1.6.3), 00:58:29/00:02:53, flags: T
  Incoming interface: FastEthernet0/0, RPF nbr 6.6.0.65
  Outgoing interface list:
    Vlan2, Forward/Sparse-Dense, 00:58:07/00:02:22

(*, 227.1.6.5), 01:13:58/00:02:50, RP 6.6.1.4, flags: S
  Incoming interface: FastEthernet0/0, RPF nbr 6.6.0.65
  Outgoing interface list:
    Vlan3, Forward/Sparse-Dense, 00:16:38/00:02:50

(6.6.0.19, 227.1.6.5), 00:16:38/00:02:20, flags: PR
  Incoming interface: FastEthernet0/0, RPF nbr 6.6.0.65
  Outgoing interface list: Null

(*, 224.0.1.39), 04:44:43/stopped, RP 0.0.0.0, flags: DC
  Incoming interface: Null, RPF nbr 0.0.0.0
  Outgoing interface list:
    Vlan4, Forward/Sparse-Dense, 04:44:10/00:00:00
    Vlan3, Forward/Sparse-Dense, 04:44:10/00:00:00
    FastEthernet0/0, Forward/Sparse-Dense, 04:44:10/00:00:00
    FastEthernet0/1, Forward/Sparse-Dense, 04:44:10/00:00:00

(6.6.1.4, 224.0.1.39), 00:10:05/00:02:54, flags: T
```

Figure 7: Show ip mroute - Full Multicast Routing Table

5.3 Verification of Failover Using Auto-RP in PIM Sparse-Dense Mode

From the Show ip pim rp mapping output displayed below (Figure 8), entered on Router 1 (the RP Mapping Agent), it is confirmed that 6.6.1.3/32 is acting as the backup RP for 6.6.1.5.

To verify the current RP for multicast streams, the commands Show ip pim rp 227.1.6.3, Show ip pim rp mapping, and debug ip pim auto-rp are entered.



To check for the current Rendezvous point (RP) for the multicast streams, the commands 'Show ip pim rp 227.1.6.3' and 'Show ip pim rp mapping' and 'debug ip pim auto-rp' are entered.

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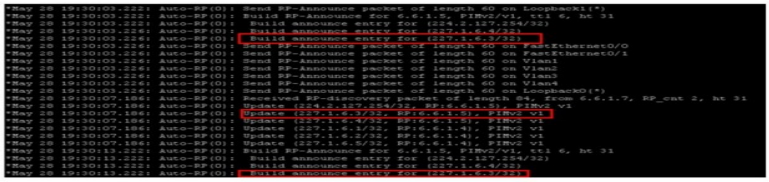
Figure 8: Show ip pim rp mapping on Router 1 (RP Mapping Agent) - Before Failover

Debug ip pim Auto-rp



To check the failover, the Loopback 0 interface with RP- address 6.6.1.5/32 is shut down using the command 'Shutdown' on the command line interface on router 3. After the shutdown, the commands 'Show ip pim rp mapping', 'show ip pim rp address 227.1.6.3' are used to verify if the failover has been activated. This means the RP address for the multicast streams 227.1.6.3, 227.1.6.4 and 224.2.127.254 would have changed from 6.6.1.5/32 on loopback 0 on router 3 (R3) to 6.6.1.3/32 on loopback 1 on router 4 which is acting as the backup.

Figure 9: Debug ip pim auto-rp - Auto-RP Debug Output

To test failover, the Loopback 0 interface with RP address 6.6.1.5/32 is shut down using the shutdown command on Router 3. After the shutdown, Show ip pim rp mapping and Show ip pim rp address 227.1.6.3 are used to confirm that failover has been activated. This confirms that the RP address for multicast streams 227.1.6.3, 227.1.6.4, and 224.2.127.254 has changed from 6.6.1.5/32 on Loopback 0 on R3 to 6.6.1.3/32 on Loopback 1 on R4, which is operating as the backup RP.

Figures 10-11: Show ip pim rp mapping and Show ip pim rp 227.1.6.3 - After Failover

Conclusion

The deployment of multicasting to deliver video streams across the data network to multiple rooms within GoodNight Plc represents an efficient and cost-effective solution compared to the traditional subscription television channel system previously in place. Beyond audio and video broadcasting, multicasting supports a broad range of applications including video conferencing, web caching, and video collaboration. The implementation of PIM Sparse-Dense Mode with load-balancing failover significantly enhances the fault tolerance of the network, ensuring uninterrupted service delivery even in the event of an RP failure.

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